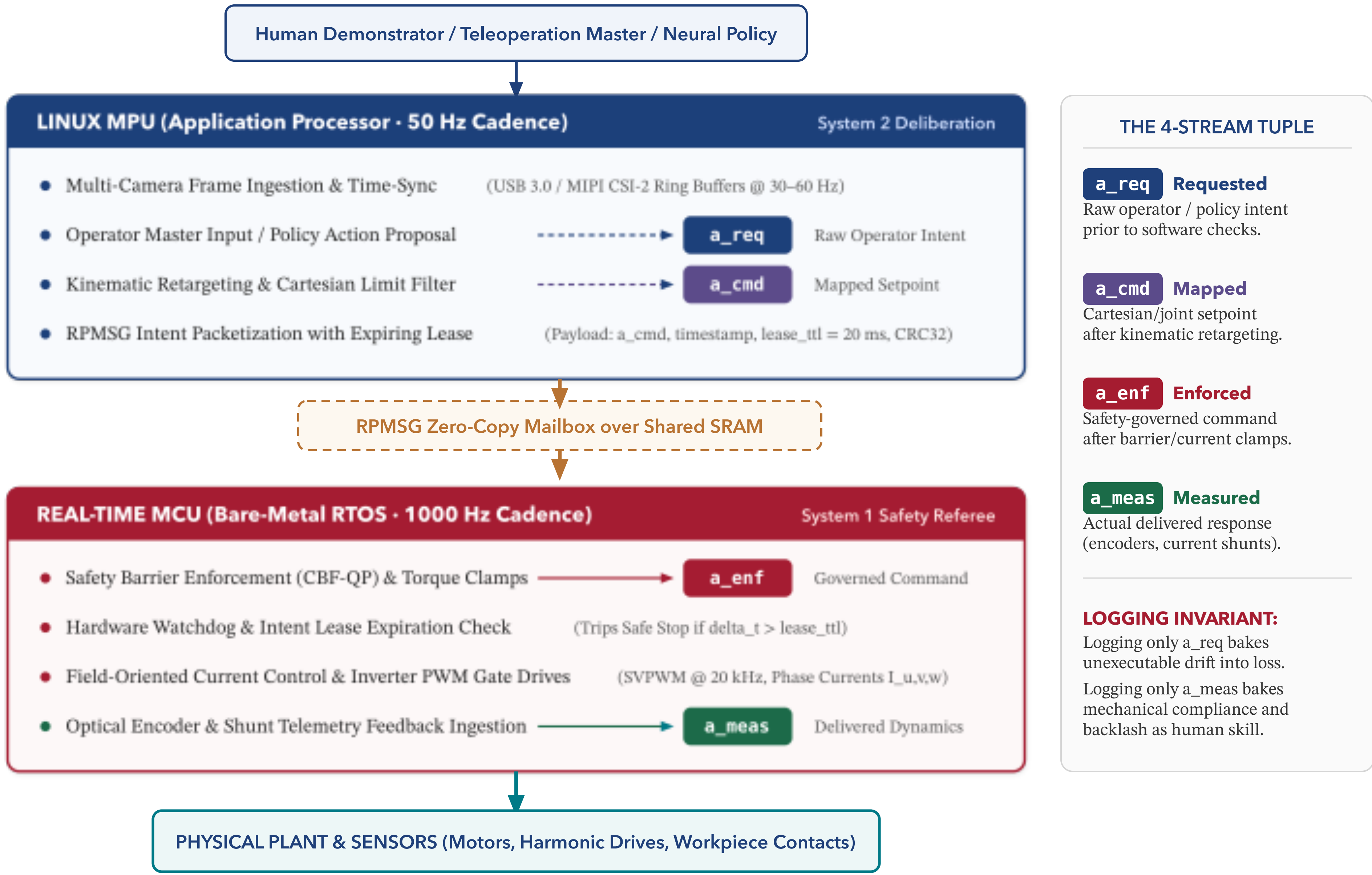
Human Demonstrator / Teleoperation Master / Neural Policy
LINUX MPU (Application Processor · 50 Hz Cadence)
System 2 Deliberation
Multi-Camera Frame Ingestion & Time-Sync
(USB 3.0 / MIPI CSI-2 Ring Buffers @ 30–60 Hz)
Operator Master Input / Policy Action Proposal
a_req
Raw Operator Intent
Kinematic Retargeting & Cartesian Limit Filter
a_cmd
Mapped Setpoint
RPMSG Intent Packetization with Expiring Lease
(Payload: a_cmd, timestamp, lease_ttl = 20 ms, CRC32)
RPMSG Zero-Copy Mailbox over Shared SRAM
REAL-TIME MCU (Bare-Metal RTOS · 1000 Hz Cadence)
System 1 Safety Referee
Safety Barrier Enforcement (CBF-QP) & Torque Clamps
a_enf
Governed Command
Hardware Watchdog & Intent Lease Expiration Check
(Trips Safe Stop if delta_t > lease_ttl)
Field-Oriented Current Control & Inverter PWM Gate Drives
(SVPWM @ 20 kHz, Phase Currents I_u,v,w)
Optical Encoder & Shunt Telemetry Feedback Ingestion
a_meas
Delivered Dynamics
PHYSICAL PLANT & SENSORS (Motors, Harmonic Drives, Workpiece Contacts)
THE 4-STREAM TUPLE
a_req Requested
Raw operator / policy intent prior to software checks.
a_cmd Mapped
Cartesian/joint setpoint after kinematic retargeting.
a_enf Enforced
Safety-governed command after barrier/current clamps.
a_meas Measured
Actual delivered response (encoders, current shunts).
LOGGING INVARIANT:
Logging only a_req bakes unexecutable drift into loss.
Logging only a_meas bakes mechanical compliance and backlash as human skill.